

Assignment 3

1. Give an example of a decreasing sequence of Borel sets $E_1 \supseteq E_2 \supseteq E_3 \dots$ such that

$$\left| \bigcap_{k=1}^{\infty} E_k \right| \neq \lim_{k \rightarrow \infty} |E_k|$$

Consider the sets: $E_k = \bigcup_{n \in \mathbb{N}} (n - \frac{1}{k}, n + \frac{1}{k})$. Each E_k is an interval of radius $\frac{2}{k}$ around every natural number. The intersection of all $E_k = \mathbb{N}$, and thus $|\bigcap_{k \in \mathbb{N}} E_k| = 0$. However, for all $k \in \mathbb{N}$, $|E_k| = \sum_{n=1}^{\infty} \frac{2}{k} = \infty$.

2. For a measure space $(X, \mathcal{S}, \mu)$ let $\{E_k\}_{k=1}^{\infty}$ be a sequence in $\mathcal{S}$ such that $\sum_{k=1}^{\infty} \mu(E_k) < \infty$. Prove that

$$\mu \left(\bigcap_{n=1}^{\infty} \left[\bigcup_{k=n}^{\infty} E_k \right] \right) = 0.$$

Proof. Notice that $A_n = \bigcup_{k=n}^{\infty} E_k$ is a decreasing sequence of sets. That is, $\mu(A_1) \geq \mu(A_2) \geq \dots$. Because $\sum_{k=1}^{\infty} \mu(E_k) < \infty$, by countable subadditivity, $\mu(\bigcup_{k=1}^{\infty} E_k) \leq \sum_{k=1}^{\infty} \mu(E_k) < \infty$. We now use continuity of measure from above on A_n to show that $\mu(\bigcap_{n=1}^{\infty} A_n) = \lim_{n \rightarrow \infty} \mu(A_n) = \lim_{n \rightarrow \infty} \mu(\bigcup_{k=n}^{\infty} E_k)$. We now only need to show that $\lim_{n \rightarrow \infty} \mu(\bigcup_{k=n}^{\infty} E_k) = 0$. We can express A_n as a disjoint union as follows:

$$A_n = \left(\bigcup_{k=n}^{\infty} E_k \setminus \bigcup_{k=n+1}^{\infty} E_k \right) \cup \left(\bigcup_{k=n+1}^{\infty} E_k \setminus \bigcup_{k=n+2}^{\infty} E_k \right) \cup \dots$$

Thus, it follows that:

$$\begin{aligned} \mu(A_n) &= \mu \left[\left(\bigcup_{k=n}^{\infty} E_k \setminus \bigcup_{k=n+1}^{\infty} E_k \right) \cup \left(\bigcup_{k=n+1}^{\infty} E_k \setminus \bigcup_{k=n+2}^{\infty} E_k \right) \cup \dots \right] \\ &= \mu \left(\bigcup_{k=n}^{\infty} E_k \right) - \mu \left(\lim_{k \rightarrow \infty} \bigcup_{k=n}^{\infty} E_k \right). \end{aligned}$$

But notice that $A_n = \bigcup_{k=n}^{\infty} E_k$, thus $\mu(A_n) - \mu(\bigcup_{k=n}^{\infty} E_k) = 0$. So finally,

$$\mu(A_n) - \mu \left(\bigcup_{k=n}^{\infty} E_k \right) = 0 = \mu \left(\lim_{k \rightarrow \infty} \bigcup_{k=n}^{\infty} E_k \right) = \lim_{n \rightarrow \infty} \mu \left(\bigcup_{k=n}^{\infty} E_k \right) \text{ (by continuity of measure)}$$

□

3. Consider the σ -algebra $\mathcal{L}$ from class, defined as

$$\mathcal{L} = \{D \subseteq \mathbb{R} : \text{For every } \epsilon > 0 \text{ there exists a closed set } F \subseteq D \text{ such that } |D \setminus F| < \epsilon\}.$$

Prove every set in $\mathcal{L}$ is Lebesgue measurable.

Proof. Let $D \in \mathcal{L}$. Thus, there must exist a closed set $F \subseteq D$ s.t. $|D \setminus F| < \epsilon$. Because this holds for all $\epsilon > 0$, it must be the case that there exists a sequence of closed subsets $F_n \subseteq D$ s.t. $\lim_{n \rightarrow \infty} |D \setminus F_n| = 0$. Thus, $\lim_{n \rightarrow \infty} F_n = F$ s.t. $|D \setminus F| = 0$. Because closed sets are Borel, this implies that there exists a Borel subset such that $|D \setminus B| = 0$, $B \in \mathcal{B}$. Therefore D must be a Lebesgue measurable set. Because D is an arbitrary set from $\mathcal{L}$, all sets in $\mathcal{L}$ must be Lebesgue measurable. $\square$

Optional Bonus: Prove that $\mathcal{L}$ is actually equal to the σ -algebra of Lebesgue measurable sets.

Proof. Let $\mathcal{L}^*$ be the set of Lebesgue measurable sets. We have already shown that $\mathcal{L} \subseteq \mathcal{L}^*$. Thus, we now must show that $\mathcal{L}^* \subseteq \mathcal{L}$. Let $D \in \mathcal{L}^*$. Thus, by definition of Lebesgue measurability, there must exist a Borel set $B \subseteq D$ s.t. $|D \setminus B| = 0$. We showed in class that given a Borel set V and $\epsilon > 0$, there exists a closed subset $F \subseteq V$ s.t. $|V \setminus F| < \epsilon$. We can now apply this result to get a set $C \subseteq B$ s.t. $|B \setminus C| < \epsilon$. Thus, $|D \setminus C| < \epsilon$. We have found a closed subset C of $D \in \mathcal{L}^*$ s.t. $|D \setminus C| < \epsilon$. Thus, $D \in \mathcal{L}$. Because D was chosen arbitrarily, every set in $\mathcal{L}^*$ must be in $\mathcal{L}$. Therefore $\mathcal{L}^* \subseteq \mathcal{L}$. We have shown that $\mathcal{L}^* \subseteq \mathcal{L}$ and $\mathcal{L} \subseteq \mathcal{L}^*$, so $\mathcal{L} = \mathcal{L}^*$. $\square$